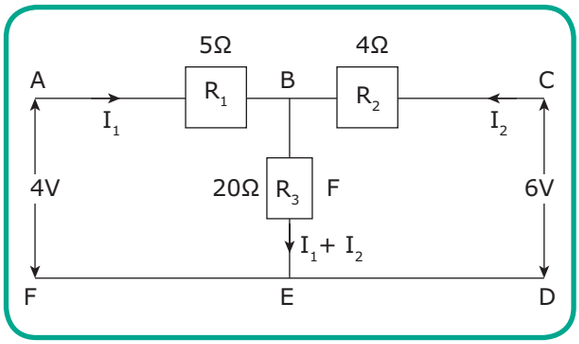


Figure 1.49

By applying Kirchhoff's second law (voltage law)

In circuit *ABEFA*,

$$5I_1+20(I_1+I_2) = 4$$

$$25I_1+20I_2 = 4 \qquad \text{Eqn.. 1}$$

In circuit CBEDC,

$$4I_2+20(I_1+I_2) = 6$$

$$20I_1+24I_2 = 6 \qquad \text{Eqn.. 2}$$

To solve this, multiply Eqn.1 by 4 on both sides, we get,

$$100I_1 + 80I_2 = 16 \qquad \text{Eqn.. 3}$$

Multiply Eqn.2 by 5 on both sides, we get,

$$100I_1 + 120I_2 = 30 \qquad \text{Eqn.. 4}$$

Subtracting Eqn.3 from 4, we get,

$$100I_1 + 120I_2 = 30$$
$$100I_1 + 80I_2 = 16 \quad (-)$$
$$40I_2 = 14$$

$$I_2 = \frac{14}{40}$$

$$I_2 = 0.35 \text{ A}$$

Substitute the value of I_2 in Eqn. 1, we get,

$$25I_1 + 20(0.35) = 4$$

$$25I_1 = 4 - 7 = -3$$

$$I_1 = \frac{-3}{25}$$

$$I_1 = -0.12 \text{ A}$$

The minus symbol shows the current flows in the opposite direction.

1.11 CAPACITOR

A component which stores electrical energy and discharges when needed is called capacitor.

The unit for measuring capacitance is Farad, which is used to measure higher values.

Other smaller value capacitors are measured using micro-Farad (μ*F*) shortly *μF*, pico-Farad (*pF*), and nano-Farad (*nF*).

The ability to store electrical energy is called as capacitance. Figure 1.50 shows the internal construction of a capacitor. The capacity to store electrical energy is termed as capacitance and the electric device to store it is termed as Capacitor. Like resistors, capacitors are also manufactured with admissible tolerance of 5% - 20%.

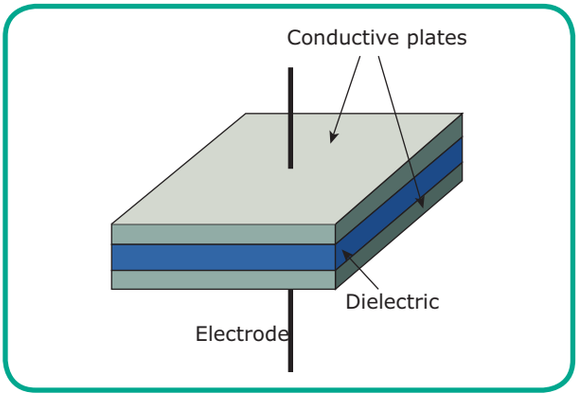


Figure 1.50 Capacitor

DO YOU KNOW? **Unit Farad was named after the British scientist Michael Faraday.**

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